

No Surprises?

Prediction Markets and the Anticipation of Macroeconomic Announcements

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Motivation

Macro announcements are expectation problems

The same official release can be a major market shock or a non-event, depending entirely on what was expected immediately before it arrived.

- CPI and labor-market releases move Treasury futures within seconds of 8:30 ET.
- FOMC decisions reprice the short-rate curve at 14:00 ET.
- In each case the asset-price move reflects the *size of the surprise*, not just the level of the outcome.

Traditional proxies for pre-release expectations such as surveys, futures, nowcasts each have limitations. None gives a real-time, full probability distribution over the official outcome.

Can prediction markets fill that gap?

What Kalshi Is

Kalshi is a CFTC-regulated prediction-market exchange, supported by market makers such as Susquehanna. It offers binary YES/NO contracts on a wide range of events.

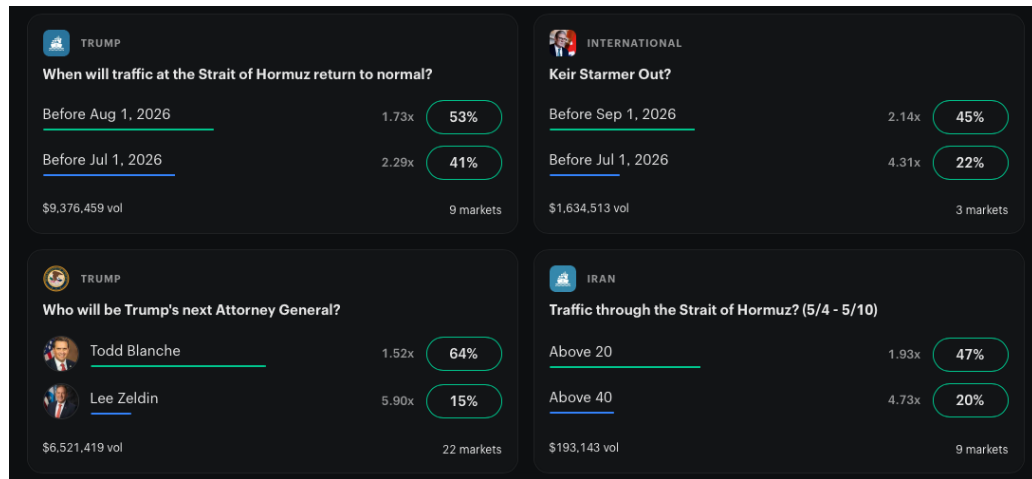
A Kalshi contract may pay one dollar if:

- headline CPI falls in a given bracket;
- nonfarm payrolls exceed a threshold;
- the unemployment rate equals a particular rounded value;
- the FOMC target range ends in a specified bracket.

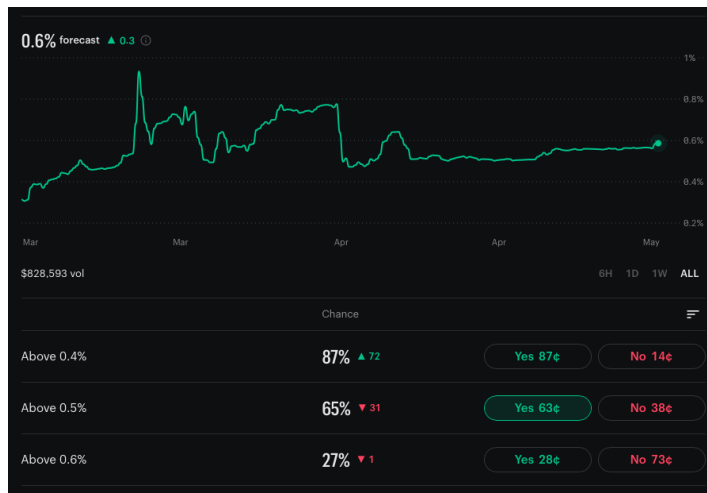
Why this is a useful object

Kalshi is not a survey and not an asset-price derivative. It is a real-money event market whose payoff is tied to the same official value that appears in the macro release. It is a natural candidate for a real-time expectation measure.

Kalshi Market Interface



Inside a (Boring) Kalshi Contract



Kalshi Contracts as Arrow–Debreu Securities

Each Kalshi YES contract on bracket B_k is an **Arrow–Debreu security** paying \$1 if $Y_e \in B_k$:

$$q_{e,k} \approx \mathbb{P}_t^K(Y_e \in B_k).$$

Normalising the full ladder $\sum_k q_{e,k} = 1$ gives a market-implied PMF. The conditional-mean forecast follows immediately:

$$\hat{Y}_e^K = \sum_k q_{e,k} m_k.$$

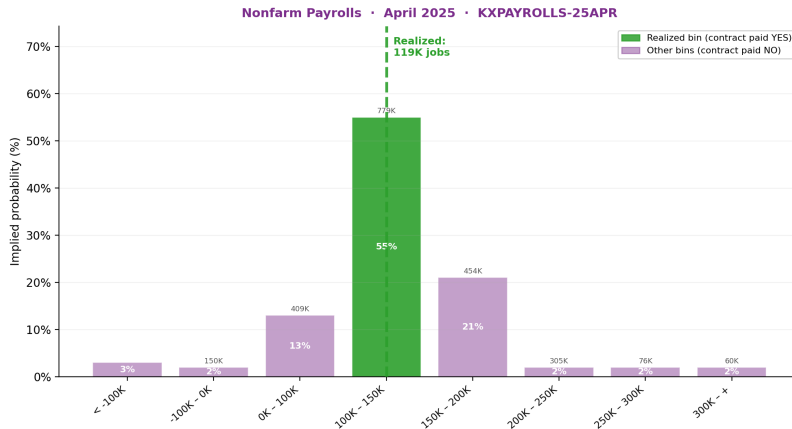
Gürkaynak & Wolfers (2006) were first to build PMFs from binary macro-derivatives.

Swanson, Wang & Wu (2025) and Diercks, Katz & Wright (2026) apply this to Kalshi and construct PMFs from Kalshi contracts.

Specifically, DKW find FOMC & CPI well-calibrated: Kalshi median matches the realised fed funds rate with zero error on the eve of every FOMC since 2022.

From Prices to Probabilities, Example: Nonfarm Payrolls

From Arrow-Debreu Prices to a Probability Distribution
Each bar = market-implied probability that payrolls land in that bracket · Green = realized outcome



Our Question

Following, we ask:

Does Kalshi deliver the same forecasting quality
across the full macro universe?

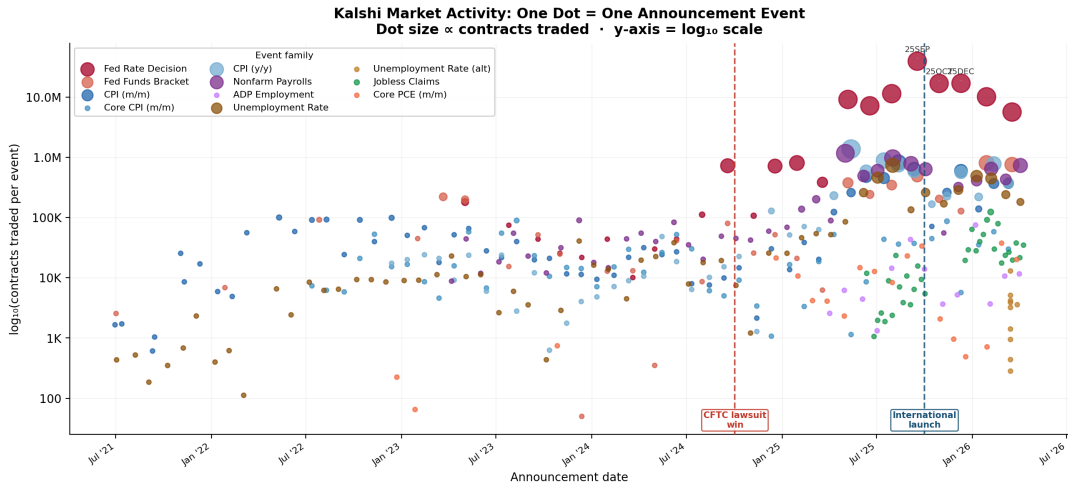
DKW focus on FOMC and CPI. We extend to labor-market announcements: **Payrolls, Jobless Claims, and Unemployment Rate.**

What We Are Predicting

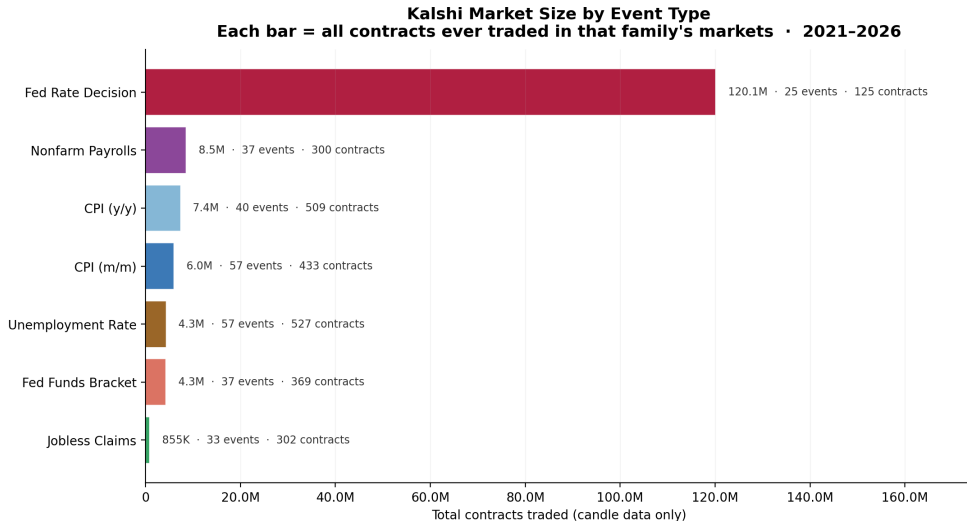
Kalshi Prediction Market — Dataset Overview All 8 macro event families · Sample: July 2021 - April 2026

Family	What it measures	Events in sample	Sample span	Median event volume	Total candle contracts
Fed Rate Decision	Fed rate change in bps (cut/hold/hike)	25	2023-2026	385K	120.1M
Fed Funds Bracket	Fed funds target upper bound (%)	37	2021-2026	486K	4.3M
CPI (m/m)	Headline CPI month-over-month (%)	57	2021-2026	28K	6.0M
Core CPI (m/m)	Core CPI month-over-month (%)	46	2022-2026	17K	1.1M
CPI (y/y)	Headline CPI year-over-year (%)	40	2022-2026	47K	7.4M
Nonfarm Payrolls	Nonfarm payrolls change (thousands)	37	2023-2026	56K	8.5M
Unemployment Rate	Unemployment rate (%)	57	2021-2026	14K	4.3M
Jobless Claims	Initial jobless claims (raw count)	33	2025-2026	19K	855K

Event Coverage



Market Depth



Data Construction

Official outcomes

- FRED/ALFRED first-release macro data.
- CPI MoM and YoY.
- Payrolls, unemployment, and jobless claims.
- FOMC target-rate outcomes.

Survey benchmarks

- Reuters/Datastream survey dispersions and medians.
- NY Fed FOMC surveys for rate-decision benchmarks.

Core Kalshi panel

Kalshi API provides detailed metadata on every resolved market: contract brackets, minute-level candlesticks, prices, volumes, open interests...

For each event, the contract ladder is collapsed into a final pre-release probability distribution and matched to the realized official outcome and survey benchmark.

First Look: Near-Hit Rates

Family	n	Near-hit band	Mean hit	Median hit
FOMC target	37	exact / $\leq 0.125\text{pp}$	37/37 (100%)	36/37 (97%)
FOMC decision	24	exact 25bp move	23/24 (96%)	23/24 (96%)
CPI YoY	36	0.05pp	17/36 (47%)	18/36 (50%)
CPI MoM	44	0.05pp	17/44 (39%)	16/44 (36%)
Unemployment	44	0.05pp	11/44 (25%)	13/44 (30%)
Payrolls	33	25K jobs	8/33 (24%)	7/33 (21%)
Jobless claims	32	5K claims	7/32 (22%)	9/32 (28%)

Descriptive check

A hit means the Kalshi mean or median lands within one strict natural rounding band of the official release. This makes the FOMC result visually obvious, but exact hit rates are too coarse for continuous releases.

Calibration Framework

Following DKW we evaluate forecasts via the Mincer–Zarnowitz regression (Mincer & Zarnowitz, 1969):

$$Y_e = \alpha + \beta \mathbb{E}_{t^-}^K[Y_e] + u_e.$$

Calibration criteria

- *MZ calibration*: joint F -test of $H_0: \alpha = 0, \beta = 1$ not rejected.
- *Economic calibration*: MZ-calibrated with meaningful fit, requiring $R^2 > 0.5$.

Benchmarks

- Survey median (Reuters/Datastream).
- Futures-implied expectation where available.
- Both evaluated under the same MZ framework.

Key identifying assumption

Snapshot prices are taken at $t_e^- = \min(\text{Kalshi close}, \tau_e) - 1 \text{ min}$, so the forecast. The ex-ante timing follows surprise measurement (Kuttner, 2001; Gürkaynak et al., 2007); the price-as-forecast interpretation follows the prediction-market literature (Wolfers & Zitzewitz, 2004).

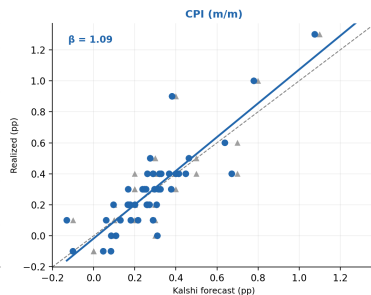
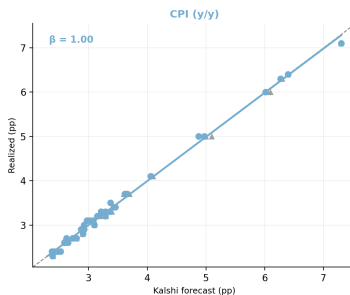
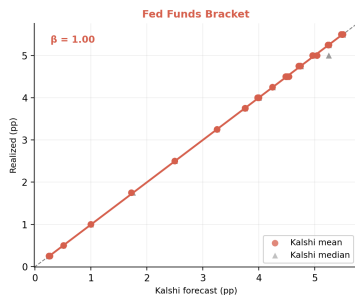
Mincer–Zarnowitz Calibration Test

Family	n	$\hat{\beta}$	R^2	MZ p	Reading
FOMC target	37	1.002	1.000	0.172	calibrated
FOMC decision	24	1.149	0.917	0.500	calibrated
CPI YoY	36	0.997	0.996	0.943	calibrated
CPI MoM	44	1.090	0.726	0.810	calibrated
Unemployment	44	0.810	0.641	0.176	calibrated
Payrolls	33	0.964	0.335	0.657	MZ pass; econ fail
Jobless claims	32	0.030	0.001	0.075	Both failed

The MZ p -value tests $H_0: \alpha = 0, \beta = 1$, so high p means we fail to reject exact calibration. We call a family “calibrated” only when MZ non-rejection is paired with meaningful fit. Payrolls is MZ-compatible, but economically weak because R^2 is low and forecast errors are large.

Calibration: Policy & Inflation

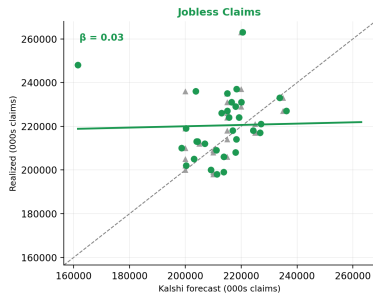
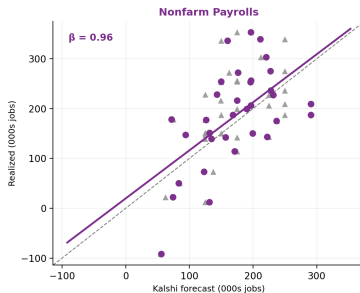
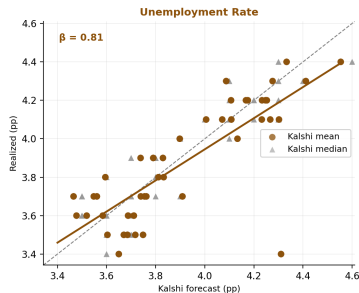
MZ Calibration: Policy & Inflation · Kalshi vs. Survey
(dashed = 45° ideal; $\hat{\beta}$ from Kalshi OLS)



Kalshi mean vs. realized outcome. Dashed = 45° ideal. $\hat{\beta}$ from Kalshi OLS. FOMC target and decision are tightly calibrated. CPI YoY is very close to the 45° line; CPI MoM is good but noisier.

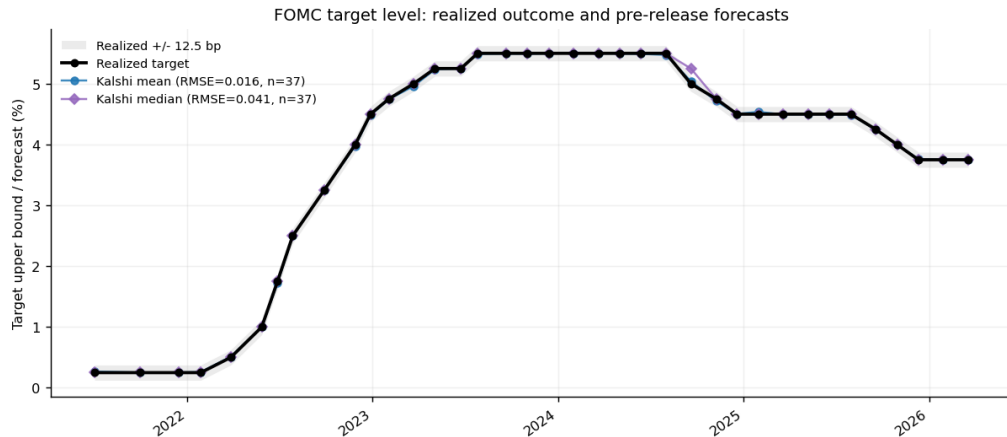
Calibration: Labor Markets

MZ Calibration: Labor Markets · Kalshi vs. Survey
(dashed = 45° ideal; β from Kalshi OLS)

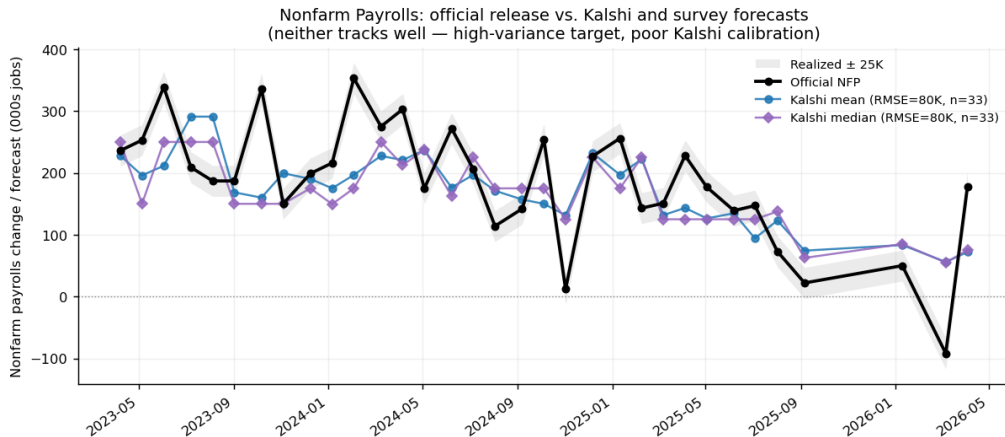


Unemployment rate is reasonable ($\hat{\beta} \approx 0.81$). Payrolls is noisy ($\hat{\beta} \approx 0.96$ but wide scatter). Jobless claims is essentially flat ($\hat{\beta} \approx 0.03$) — the market fails to track the outcome.

Forecast Competence: The Good



Forecast Competence: The Bad



Kalshi horizon-error path for nonfarm payrolls.

The Labor-Market Puzzle

Event type	Kalshi $\hat{\beta}$	Calibrated?	Mean hit
Unemployment	≈ 0.81	Yes	25%
Payrolls	≈ 0.96	No	24%
Jobless claims	≈ 0.03	No	22%

The labor-market puzzle

Labor is not uniformly bad. Unemployment is calibrated, plausibly because it is slow-moving and changes little between releases. Payrolls and claims are the high-variance cases: payrolls has a roughly unit slope but large errors and weak fit, while jobless claims is nearly flat.

Why Does Kalshi Fail for Labor Markets?

Three diagnostic tests

- ① **Horizon errors:** does the market learn as release day approaches?
- ② **Round-number salience:** does probability mass pile up at salient thresholds?
- ③ **Liquidity screens:** do forecasts improve when the market has real open interest?

The answer

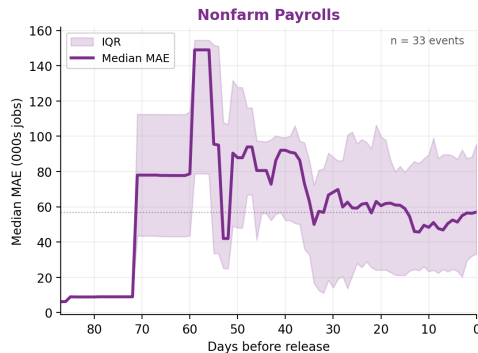
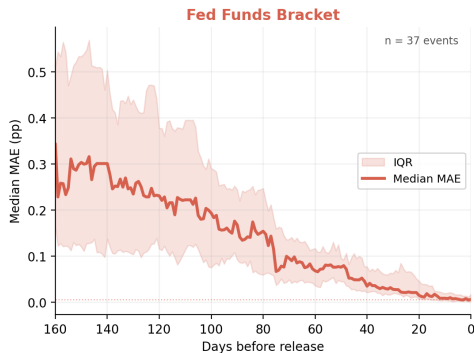
Labor failures have three pieces: weak pre-release learning, salient contract thresholds, and limited liquidity. Filtering to liquid markets helps substantially, but does not fully rescue payrolls or jobless claims.

Horizon Errors

Question. Do prediction markets absorb information before the scheduled release, or does the final surprise remain intact until announcement time?

Estimate. Track $MAE_f(h) = \text{median}_{e \in f} |Y_e - \hat{\mu}_{e,-h}^K|$ from $t - 30$ to release.

Forecast Error Decay: FOMC vs. Nonfarm Payrolls



Horizon Error Test Results

Event type	Unit	$t - 30$	$t - 7$	$t - 1$	Release	Absorbed
FOMC target	pp	0.028	0.007	0.005	0.006	78%
FOMC decision	bps	3.96	0.74	0.43	0.25	94%
CPI YoY	pp	0.125	0.076	0.048	0.053	58%
CPI MoM	pp	0.147	0.070	0.069	0.072	51%
Unemployment	pp	0.120	0.112	0.122	0.105	13%
Payrolls	K jobs	68.3	46.9	56.3	57.0	17%
Jobless claims	K claims	–	9.47	9.24	9.51	-0.4% [†]

Entries are median absolute forecast errors from the Kalshi mean/consensus path. “Absorbed” is the percentage reduction in error from $t - 30$ to release. [†]For jobless claims, the cell uses $t - 7$ to release.

Round-Number Saliency

Question. Within a given event, do bins that straddle a round-number threshold attract excess probability mass beyond what their distance from the Kalshi mean predicts?

Regression. (event fixed effects, b indexes PMF bins within event e):

$$q_{e,b} = \mu_e + \beta \mathbf{1}[\text{salient}_b] + \gamma_1 |d_{e,b}| + \gamma_2 |d_{e,b}|^2 + \varepsilon_{e,b},$$

where $q_{e,b}$ is the bin probability in percentage points, μ_e is the event FE, and $d_{e,b} = (m_b - \hat{\mu}_e)/\hat{\sigma}_e$ is distance from the Kalshi mean in event-SD units, ensuring we do not confuse saliency with the modal bin mechanically having higher probability.

Salience Test Results

Family	Major salient	Secondary salient	Major coef. (q)	Secondary coef. (q)
FOMC target	whole pp	half pp	-3.89 (0.542)	+2.68 (0.792)
CPI YoY	whole pp	half pp	-0.33 (0.915)	-1.14 (0.542)
CPI MoM	0.5 pp	0.25 pp	-0.26 (0.895)	-0.26 (0.895)
Unemp.	whole pp	half pp	-1.61 (0.478)	+0.50 (0.820)
Payrolls	100K jobs	50K jobs	+3.83 (0.115)	+4.50 (0.172)
Jobless c.	100K claims	50K claims	+12.26 (0.0003)	+6.70 (0.001)

Coefficients are extra PMF mass on salient bins, in percentage points, after event fixed effects and distance-from-mean controls. q is the Benjamini–Hochberg-corrected p -value for the null of no salience effect.

Liquidity

Question. Are labor-market prediction failures just a thin-market problem?

Test. Re-estimate the Mincer–Zarnowitz calibration regression after keeping only events with high pre-release open interest, $OI_e = \sum_{m \in e} OI_{m,t^-}$. The table compares the full sample with the top 50% of events by OI_e .

Family	Screen	n	RMSE	Improvement	$\hat{\beta}$	Calibrated?
Payrolls	all events	33	0.722	–	0.96	No
Payrolls	top 50% OI	16	0.477	34%	0.97	No
Jobless claims	all events	32	0.956	–	0.03	No
Jobless claims	top 50% OI	13	0.491	49%	2.38	No

Shortcomings and Interpretation Caveats

Market-price interpretation

- Kalshi prices are not automatically physical probabilities
- price-probability link depends on agent heterogeneity on risk aversion, wealth constraints etc. (Gjerstad, 2004; Manski, 2006; Bürgi et al., 2026)

Empirical limitations

- Total # of events are still small, especially after splitting by event family
- Kalshi traders are self-selected
- Liquidity varies sharply across contracts and over time

No Surprises?

Bottom Line

Prediction markets do not eliminate macro announcement surprises uniformly. They work best when the object is discrete, telegraphed, and low-dimensional.

Anticipated events

- FOMC target outcomes are close to ideal: well-calibrated and mostly absorbed before release.
- CPI forecasts are reasonably informative, especially relative to high-variance labor releases.

Remaining surprises

- Unemployment is reasonably calibrated, but still low hit rates.
- Payrolls and jobless claims remain difficult even in more liquid subsamples.

So the answer is: no surprises for some markets, still real surprises for others.

Further Research

Extensions

- If order-flow becomes available, test whether labor failures stem from noise-trader dominance or genuine absence of informed participation.
- Cross-platform comparisons of (e.g. Kalshi vs. Polymarket) for the same releases to separate market-structure effects from event-difficulty effects.

Prediction markets beyond forecasting

- Vast possibilities for political economic research. (e.g., Aktuğ & Torul, 2025).
- Vast possibilities for behavioral research.
- Natural experimental settings on structural breaks with CFTC lawsuit and international launch.

Thank You

M. Kerem Yavuz